

Thangasamy G

MLOps Expert

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AWS Certified Machine Learning Specialty

Technical Skills

Gen AI: OpenAI ChatGPT-4, Hugging Face Transformers, OpenSearch, LLM, Langchain, RAG, PyTorch, TensorFlow.

Cloud: AWS Sage maker, Azure Devops (ML Studio, Azure Data Lake, Azure Data Factory), Terraform, Docker, Kubernetes.

Programming Skills: Python (NumPy, Pandas, Matplotlib, Sklearn, SciPy, Seaborn), Shell Script.

ML/Data Framework: MLFlow, Airflow.

Data Collection: ELK, APIs.

CI/CD: GitHub actions, Jenkins, Jira, JFrog, SonarQube, AWS Devops, DagsHub.

Data Visualization: Matplotlib, Seaborn.

Monitoring: New Relic, Prometheus, Tableau.

Conversational Site Search

Build our own site search using Gen AI (LLM) for customers. In this method, using an advanced Artificial Intelligence system (NLP) designed to understand and respond to user queries in a natural, conversational manner.

Competitor Products Intel

Competitor products analysis using AI involves the application of artificial intelligence technologies to gather, analyse and derive insights from data about Competitor’s products.

Single Line Diagram

Creating Single line diagrams (SLDs) using AI involves leveraging artificial intelligence technologies to automate and optimize the process of generating electrical schematic diagrams.

Environments: AWS, Azure, Open Search (Vector DB), LLM, Langchain, OpenAI, GitHub Actions, Terraform.

AI/ML: AWS Sage maker, Azure ML Studio, APIs, Azure Kubernetes Service, DagsHub.

Data Processing: MLFlow, Airflow, Pandas, NumPy, Sklearn, NumPy, Matplotlib, Seaborn.

Generative AI: ChatGPT-4/GPT-3.5 (OpenAI).

DL Framework: PyTorch

ML Algorithms: Yolov5, RAG, Linear & Logistic Regression, KNN, Information Retrieval (BM25), Universal Sentence Encoder, Brisque, BERT, GPT, Fast Text, CLIP.

Summary

With over 12 years of diverse expertise in Artificial Intelligence (AI), Machine Learning Operations (MLOps), Gen AI, Devops, AWS, Azure and Linux System Administration. I bring a comprehensive skill set honed through hands-on experience and a commitment to innovation. My journey has seen me adeptly navigate the cutting-edge realms of AI, Gen AI and ML leveraging advanced techniques to develop intelligent solutions that drive business success.

Work Experience

MLOPS Expert, Schneider Electric May 2023 to Oct 2024

- Manage a team of data engineers and developers in creating AI/ML applications that increased operational efficiency by 25%.
- Developed and deployed a Retrieval-Augmented Generation (RAG) pipeline leveraging LLMs to improve the accuracy and relevance of AI generated content by integrating external knowledge sources.
- Implemented RAG strategies to optimize the performance of LLMs in real-time applications, resulting in a 25% increase in the accuracy of generated responses in customer support scenarios.
- Experienced with integrating LLMs into existing knowledge management systems to enable contextually accurate generative AI solutions.
- Architected and implemented scalable ML pipelines using Kubernetes and Docker, reducing model deployment time by 40%.

MLOps Engineer, Wells Fargo, Bengaluru

Aug 2020 to May 2023

- Designed and implemented MLOps pipelines for various financial use cases, including credit risk assessment, fraud detection, clinical health care analysis and customer churn prediction.
- Utilized Azure Machine Learning for model training, retraining, ML pipeline deployment and management.
- Leveraged Azure Databricks for large scale data processing and feature engineering.
- Integrated ML pipelines with CI/CD workflows using Azure DevOps for continuous integration and deployment.
- Monitored and optimized ML models in production to ensure performance and drift detection.
- Collaborated with data scientists and quants to understand their needs and translate them into operational MLOps solutions.
- Communicated effectively with business stakeholders on the value and impact of ML models.

Smart Consumer 360 Predictor

- Developed a model to predict consumer behaviour in online shopping for an E-Commerce environment.
- Data Mining, Data Aggregation, Data Transformation, Data preparation, Data analysis and wrangling.
- Used Decision trees, Mathematical calculations, classification and feature correlations, boosting to train classifiers.
- EDA on features that influenced purchase behaviour.
- Accuracy achieved: 87%

EDA and Visualization (Banking & Finance)

- Performed both advanced qualitative and quantitative analysis of high-volume data bases, to identify trends, patterns and correlations that could improve overall business performance.
- Provide data backed business insight for better decision making using large data set and bringing out insight from data.
- Developed dashboards and analytics report using Python, Jupyter notebook to enable review decision making and continuously engaged with functional teams to support decision making.

Business Analysis /Price prediction

- Extracting and Managing data from online and field survey to convert into meaningful information.
- Performed data cleaning, normalization using Azure ML Studio.
- Used Linear Regression for model development, Model deployment and evaluation.
- Lead a team of data analysts to deliver multiple analytics projects within set timeline.

Cloud Environments: Azure DevOps, Azure ML Studio, Azure Data Lake & Data Factory, Azure SQL DB, Azure Kubernetes Service, Jupyter Notebook.

DevOps: Docker, Kubernetes, Jenkins, GitLab, SonarQube, Terraform, JFrog, Jira, New Relic.

MLOps Tools: Python Libraries (NumPy, Sklearn, SciPy, Pandas, TensorFlow Framework, MLFlow, Matplotlib, Airflow, Seaborn) Jupyter Notebook, LLM (RAG), Langchain, Tableau, Hugging Face Transformers.

Senior Devops Engineer, L&T Technology Services, Bengaluru

Aug 2018 to June 2020

Build Managing for FCA and FORD ADAS Projects-Aptiv (Customer Location: Poland & India)

- Experience implementing Azure services such as Azure Active Directory, Azure storage, Azure Cloud services, IIS, Azure Resource Manager (ARM), Azure Blob Storage, Azure VM, Azure SQL Database, Azure Functions, Azure Service Fabric and Azure Service Bus.
- Developed ADF pipelines to move data from on-premises source systems to COSMOS, from COSMOS (with data transformation) to Azure Warehouse (staging), from Azure warehouse to Azure ML (for scoring) and appending scores back to the data in Azure warehouse.
- Involved in automating end to end model refresh process using COSMOS, Azure Data Factory (ADF) pipelines, Azure ML, Python and PowerShell scripting. Developed, maintained and provided the team with various Azure DevOps related tools like deployment tools, staged virtual environments and provisioning scripts.
- Working knowledge in deploying CI/CD system using Azure DevOps on Kubernetes container environment and for the runtime environment of CI/CD system to build, test and deployment. we have utilized Azure Kubernetes Services and Docker.
- Structured cluster Autoscaler for Azure Kubernetes Service by using Terraform and worked with scheduling, deploying and managing pods and replicas in AKS.
- Implemented comprehensive monitoring and logging solutions with Azure Monitor and Log Analytics, enhancing system reliability and uptime.

Devops Engineer, Usha Martin Technologies, Bengaluru

March 2017 to June 2018

- Implemented CI/CD pipelines for multiple projects using Jenkins, SonarQube, Coverity, GitLab, Terraform and AWS Code Pipeline, reducing manual intervention and improving release frequency.
- Managed AWS resources and services, including EC2, S3, RDS, VPC, IAM, Lambda and Route 53 for various development and production environment.

- Utilized Docker for containerization and Amazon ECS for container orchestration, enabling seamless deployment and scaling of microservices.
- Configured and maintained monitoring solutions with AWS CloudWatch, ELK Stack and Prometheus to monitor system performance and application metrics.
- Automated infrastructure provisioning and configuration management tasks using Terraform improving deployment consistency and reliability.
- Ensured security best practices by implementing IAM policies, Security groups, VPC configurations and Enhanced system security by conducting regular audits and implementing necessary security measures.

Cloud Environment: AWS DevOps (AWS Code Pipeline, EC2, S3, RDS, Lambda, IAM, CloudFormation, CloudWatch, Route53, ECS etc...)

DevOps: Docker, Kubernetes, Jenkins, SonarQube, Coverity, GitLab, Terraform, New Relic, Prometheus.

Programming Skills: Python, Shell.

Devops Engineer, Nixserv IT Solutions Pvt Limited, Bengaluru

Sep 2013 to March 2017

- Designed, implemented, and managed CI/CD pipelines for multiple projects, using Jenkins, GitLab CI/CD to automate build, test and deployment processes.
- Leveraged Docker and Kubernetes to containerize applications and orchestrate deployments, improving scalability and resource utilization.
- Implemented Infrastructure as Code (IaC) using Terraform to provision and manage AWS resources, reducing manual errors and increasing efficiency.
- Configured monitoring and alerting systems using Prometheus, Grafana and ELK Stack to monitor system performance and application metrics.
- Automated routine tasks and configuration management using Ansible, reducing deployment time and ensuring consistency across environments.
- Collaborated with development and operations teams to troubleshoot issues, optimize performance and implement best practices for DevOps processes.

Linux Administrator, Bharat Electronics Limited, Bengaluru

July 2012 to July 2013

- Installed, configured and maintained Linux servers (RHEL, CentOS, Ubuntu, Debian) for various applications and services, ensuring high availability and performance.
- Managed user accounts, permissions and file systems, implementing security best practices and access controls.
- Automated routine tasks and system administration processes using shell scripts (Bash, Python), cron jobs and configuration management tools.
- Monitored system performance and health using Nagios proactively identifying and resolving issues to minimize downtime.
- Implemented and maintained virtualization solutions with VMware, VirtualBox and KVM optimizing resource utilization and scalability.
- Conducted regular security audits, vulnerability assessments and patch management to protect Linux systems from threats and vulnerabilities.

Education:

- **BE-Muthayammal Engineering College, Anna University, Apr 2011.**
Dissertation: “**Electronics and Communication Engineering**”

Certification:

- Achieved certification from the **AWS Certified Machine Learning - Specialty**, offered by AWS.

Declaration

I hereby declare that the particulars of information and facts stated herein above are true, correct and complete to best of my knowledge and belief.

Place:

Date:

(Thangasamy Ganesan)